COSMIC UNIVERSALISM COMPUTATIONAL INTELLIGENCE INITIATIVE

Cosmic Universalism Website

Cosmic Breath Interactive Cycle Explorer

Implementation and Operations Manual

A controlled specification for a cinematic, accessible, educational explorer with free pointer dragging, intelligent TOM snapping, settle-to-details behavior, adaptive high-fidelity SVG/Canvas graphics, explicit reset controls, and an unchanged structural overview whenever exploration closes.

Repository	CosmicUniversalismStatement
Protected branch	`main`
Stable website branch	`website`
Live integration HEAD	`d426b78`
Latest tagged release	`website-v1.3-aurelius-release` at `5a56085`
Proposed feature branch	`feature/cosmic-breath-cycle-explorer` (not created)
Target route	`/CosmicUniversalismStatement/cosmic-breath/`
Technology	Astro, TypeScript, semantic HTML, modern CSS, SVG, Vitest, lightweight browser JavaScript
Feature scale	51 selectable TOM states, free planar drag, snap/settle detail reveal, overview, and next-breath reset transition
Document version	1.1
Verified	July 22, 2026

William Maddock https://github.com/willmaddock/CosmicUniversalismStatement

Document Control

Document	Cosmic Universalism Website: Cosmic Breath Interactive Cycle Explorer Implementation and Operations Manual
Version	1.1
Classification	Feature technical specification, interaction design standard, implementation plan, test plan, and release runbook
Repository	CosmicUniversalismStatement
Local repository	/Users/cosmos/Documents/GitHub/CosmicUniversalismStatement
Protected branch	`main`
Stable integration branch	`website`
Verified live integration HEAD	`d426b78` (full: `d426b78750f6be26eff4c8fffba954c1a7ec8aa9`)
Latest tagged application release	`website-v1.3-aurelius-release` at 5a56085
Proposed feature branch	`feature/cosmic-breath-cycle-explorer`; not created by this manual-authoring phase
Target route	/CosmicUniversalismStatement/cosmic-breath/
Primary current page	website/src/pages/cosmic-breath.astro
Shared static diagram	website/src/components/CosmicBreathDiagram.astro
Technical baseline verified	July 22, 2026
Manual package location	Outside the Git repository; no website or Git state changed

Purpose

Define the complete visitor experience, content boundaries, data model, advanced drag-and-snap system, settle-triggered educational details, hybrid SVG/Canvas graphics architecture, adaptive quality and performance budgets, accessibility behavior, implementation file scope, testing gate, and controlled release path for the Cosmic Breath Interactive Cycle Explorer.

Safety Rule

This manual authorizes no website edit and no Git-changing or deployment action. It records a future implementation specification. Before implementation, re-run the repository preflight, obtain explicit authorization for exact files, and complete the canonical TOM ledger review before presenting disputed duration values as approved.

Authority and Source Priority

When sources disagree, use this priority: (1) live repository state verified at the time of implementation; (2) the Master Implementation and Operations Manual; (3) exact current source files; (4) a separately approved canonical TOM ledger; (5) uploaded CU research files as source material; and (6) this feature manual's design recommendations. Research-source ambiguities are recorded rather than silently reconciled.

Contents

Document Control 1

START HERE - Initialize the Feature Project Chat 5

1 Part 1 - Governance and Verified Baseline 6
- 1.1 Verified repository state at manual creation . . . 6
- 1.2 Live state versus release baseline . . . 6
- 1.3 Phase authorization . . . 7

2 Part 2 - Feature Purpose, Outcome, and Protected Meaning 7
- 2.1 Public and technical identity . . . 7
- 2.2 Current baseline . . . 7
- 2.3 Protected meaning . . . 7

3 Part 3 - Visitor Experience and Interaction States 8
- 3.1 State model . . . 8
- 3.2 Overview state . . . 8
- 3.3 Free Drag and Settle-to-Details . . . 10
- 3.4 Persistent Exploration . . . 10
- 3.5 Temporary Preview Option . . . 10
- 3.6 Phase boundary behavior . . . 10
- 3.7 Return behavior . . . 10

4 Part 4 - Information Architecture and Educational Detail 11
- 4.1 Selected TOM panel . . . 11
- 4.2 Detail reveal timing . . . 11
- 4.3 Collapsed versus complete layer display . . . 12
- 4.4 Full layer drawer . . . 12
- 4.5 Source and interpretation disclosure . . . 12

5 Part 5 - Graphics, Color, Motion, and Responsive Design 12
- 5.1 Visual objective . . . 12
- 5.2 Phase palette . . . 13
- 5.3 Hybrid rendering architecture . . . 13
- 5.4 SVG foreground layers . . . 14
- 5.5 Motion rules . . . 14
- 5.6 Advanced visual effects . . . 15
- 5.7 Adaptive visual quality and performance budget . . . 15
- 5.8 Responsive layouts . . . 15
- 5.9 Annotated planning reference . . . 15

6 Part 6 - Canonical TOM Ledger Review Gate 15
- 6.1 What the uploaded sources establish . . . 16
- 6.2 Why a review gate is mandatory . . . 16
- 6.3 Required review outputs . . . 17
- 6.4 Typed approval fields . . . 17

6.5 Approval rule for production display . . . 18

7 Part 7 - Technical Architecture and File Responsibilities 18

7.1 Architecture decision . . . 18
7.2 Smallest safe file set . . . 18
7.3 Protected existing files . . . 19
7.4 Progressive enhancement . . . 19
7.5 Component scoping . . . 19

8 Part 8 - Interaction Engineering 19

8.1 Canonical controls . . . 19
8.2 Draggable subjects and free movement . . . 20
8.3 Pointer lifecycle . . . 20
8.4 Snap and settle behavior . . . 20
8.5 Reset semantics . . . 20
8.6 Native slider synchronization . . . 21
8.7 Keyboard behavior . . . 21
8.8 Marker identity rules . . . 21
8.9 Boundary guards . . . 21

9 Part 9 - Accessibility, Usability, and Content Safety 21

9.1 Accessibility baseline . . . 21
9.2 Cognitive load . . . 22
9.3 Reality and claims boundary . . . 22
9.4 Privacy and persistence . . . 22

10 Part 10 - Automated Tests, Browser Matrix, and Acceptance 22

10.1 Focused automated tests . . . 22
10.2 Full project gate . . . 23
10.3 Manual browser matrix . . . 23
10.4 Acceptance criteria . . . 23

11 Part 11 - Controlled Implementation, Git, Release, and Recovery 24

11.1 Pre-implementation preflight . . . 24
11.2 Proposed branch and release . . . 24
11.3 Bounded implementation authorization template . . . 25
11.4 Release gate . . . 25
11.5 Recovery . . . 25

12 Part 12 - Implementation Phases 26

12.1 Phase 0 - Canonical ledger approval . . . 26
12.2 Phase 1 - Typed registry and pure state engine . . . 26
12.3 Phase 2 - Static route-specific shell . . . 26
12.4 Phase 3 - Native controls and details . . . 26
12.5 Phase 4 - Pure drag, snap, and settle engine . . . 26
12.6 Phase 5 - Pointer and SVG interaction . . . 26
12.7 Phase 6 - Canvas atmosphere and adaptive quality . . . 26
12.8 Phase 7 - Full verification . . . 26

13 Part 13 - Risk Register and Decision Rules **26**

14 Appendix A - Feature Planning Worksheet **27**

15 Appendix B - Provisional Full-Cycle Source Ledger **28**
15.1 Expansion source ledger - 26 states 28
15.2 Compression source ledger - 25 states 29

16 Appendix C - Canonical Ledger Review Worksheet **29**

17 Appendix D - Browser Review Script **30**

18 Appendix E - Source Provenance **30**

19 Appendix F - Current-State Handoff Summary **31**

20 Appendix G - Revision Record **31**

START HERE - Initialize the Feature Project Chat

Successful Result

This manual is complete enough to initialize planning, ledger review, implementation, testing, and release work for the feature. Exact production duration values remain gated by the canonical TOM ledger review described in Part 6.

Required reading order

1. Document Control and source priority.
2. Part 1: governance and verified baseline.
3. Part 2: product outcome and protected meaning.
4. Part 3: visitor experience and interaction states.
5. Part 6: canonical TOM ledger review gate.
6. Parts 7-10: architecture, implementation, accessibility, and tests.
7. Part 11 before any Git or release action.

Copy-and-Paste Prompt: Initialize the Cosmic Breath Cycle Explorer Work

```
We are continuing controlled work on the Cosmic Universalism website.
Read the Master Implementation and Operations Manual and the Cosmic
Breath Interactive Cycle Explorer Implementation and Operations Manual.

Repository:
/Users/cosmos/Documents/GitHub/CosmicUniversalismStatement

Protected branch:
main

Stable website branch:
website

Recorded live integration HEAD at manual creation:
d426b78

Latest tagged application release:
website-v1.3-aurelius-release at 5a56085

Proposed feature branch:
feature/cosmic-breath-cycle-explorer

Target route:
/CosmicUniversalismStatement/cosmic-breath/
```

```
Before editing:
1. Verify repository root, origin, active branch, HEAD, tags, and status.
2. Distinguish the current live state from this manual's recorded baseline.
3. Inspect only the authorized feature files.
4. Confirm the canonical TOM ledger decision and exact approved values.
5. Report the smallest safe implementation plan.

Do not edit during the first inspection.
Do not alter main.
Do not reset, clean, stash, discard, pull, switch branches, commit,
push, merge, tag, release, deploy, add dependencies, or change CU-Time
mathematics without separate explicit authorization.
```

1 Part 1 - Governance and Verified Baseline

1.1 Verified repository state at manual creation

The user-provided Git preflight established the following current live state on July 22, 2026:

- repository root: /Users/cosmos/Documents/GitHub/CosmicUniversalismStatement;
- origin: https://github.com/willmaddock/CosmicUniversalismStatement.git;
- active branch: `website`;
- live HEAD: `d426b78`;
- remote relationship: `website` synchronized with `origin/website`;
- working tree: clean;
- tag at live HEAD: none.

The live HEAD descends from `5a56085`. Commit `d426b78` merges `b6604e1`, "Add unified website master implementation manual." Therefore, the post-v1.3 difference is documented rather than unexplained.

1.2 Live state versus release baseline

Classification	State
Current live integration branch	`website` at `d426b78`
Remote state	Synchronized with `origin/website` at inspection time
Latest tagged application release	`website-v1.3-aurelius-release` at `5a56085`
Post-release change	Unified master-documentation merge
Proposed implementation branch	`feature/cosmic-breath-cycle-explorer`; not created
Recovery point	`website-v1.3-aurelius-release`

1.3 Phase authorization

Every future implementation phase uses four gates: Inspect, Plan, Implement, and Verify. Manual creation has completed only the design and planning artifact. It does not authorize branch creation or code changes.

Do Not Continue Until Resolved

At implementation time, stop if the live branch, HEAD, tag relationship, or working tree differs from the authorized task. Never use reset, clean, stash, discard, or branch switching to conceal an unexplained difference.

2 Part 2 - Feature Purpose, Outcome, and Protected Meaning

2.1 Public and technical identity

Public name	Cosmic Breath Interactive Cycle Explorer
Technical name	Route-specific TOM cycle state explorer and structural-scale visualization
Target route	/CosmicUniversalismStatement/cosmic-breath/
Primary user outcome	Explore every expansion and compression TOM, inspect educational details and reviewed durations, move a simulated cosmic state through the complete cycle, and return to the unchanged original overview.

2.2 Current baseline

The current Cosmic Breath page renders a static accessible SVG with five labeled structural positions: sub-ztom, atom, btom, ctom, and ztom. A cyan “You Are Here / Observable Universe” marker is placed inside ctom. The page states that sub-ctom and ctom are not interchangeable, that the diagram is structural rather than physically scaled, and that the marker placement is a CU theoretical proposition rather than an empirical measurement.

The same static component is also used by the homepage hero. The implementation must not make the homepage interactive or change its baseline presentation.

2.3 Protected meaning

The feature shall preserve all of the following:

- The original overview places the present Observable Universe marker only within ctom.
- sub-ctom and ctom remain separate terms.
- The diagram is not a scale drawing, observational map, or empirical measurement.
- The detailed traveling marker is labeled **Simulated Cosmic State** outside the present ctom overview.
- CU framework statements remain classified as CU theoretical propositions or philosophical interpretation as appropriate.
- No CU-Time converter mathematics, source URL, CUCII claim boundary, or homepage behavior is altered.
- The original five-shell overview remains available without JavaScript.

Checkpoint

The explorer teaches the CU cycle. It does not claim that a visitor literally pushes the empirical Observable Universe through measured physical shells. The UI must make the simulation boundary visible in text, not only through color or animation.

3 Part 3 - Visitor Experience and Interaction States

3.1 State model

The feature contains nine top-level interaction states. Expansion and compression are phase attributes carried by the selected TOM rather than separate interaction modes.

1. **Overview**: unchanged five-shell diagram and present marker inside ctom.
2. **Explorer Ready**: all 51 TOM states are available, but no object is moving.
3. **Dragging**: a marker, TOM layer handle, or cycle handle follows mouse, finger, or stylus movement with a compact live preview.
4. **Snap Preview**: the nearest valid TOM is emphasized while the pointer is still down.
5. **Settling**: after release, the moved object eases to the canonical TOM position.
6. **Details Revealed**: the educational panel opens only after settling completes.
7. **Atom Boundary**: the explorer pauses at the end of expansion and requires deliberate continuation into btom.
8. **ZTOM Boundary**: the explorer pauses at the universal reset boundary.
9. **Next-Breath Reset**: a deliberate final inward transition from ztom to the next sub-ztom seed, followed by either continued exploration or return to overview.

At any point, a new drag interrupts the current details state, returns the interface to a lightweight motion preview, and prepares a new snap/settle cycle. Reset, Escape, and page refresh restore Overview.

3.2 Overview state

The default page shall render the current structural figure, current accessible title and description, present marker within ctom, and existing explanatory text. The explorer controls may be visible as an enhancement invitation, but the full TOM timeline remains collapsed.

Recommended entry points:

- an **Explore the Full Cosmic Breath** button below the figure;
- selecting the central sub-ztom seed with pointer or keyboard;
- a text link near “Reading the diagram.”

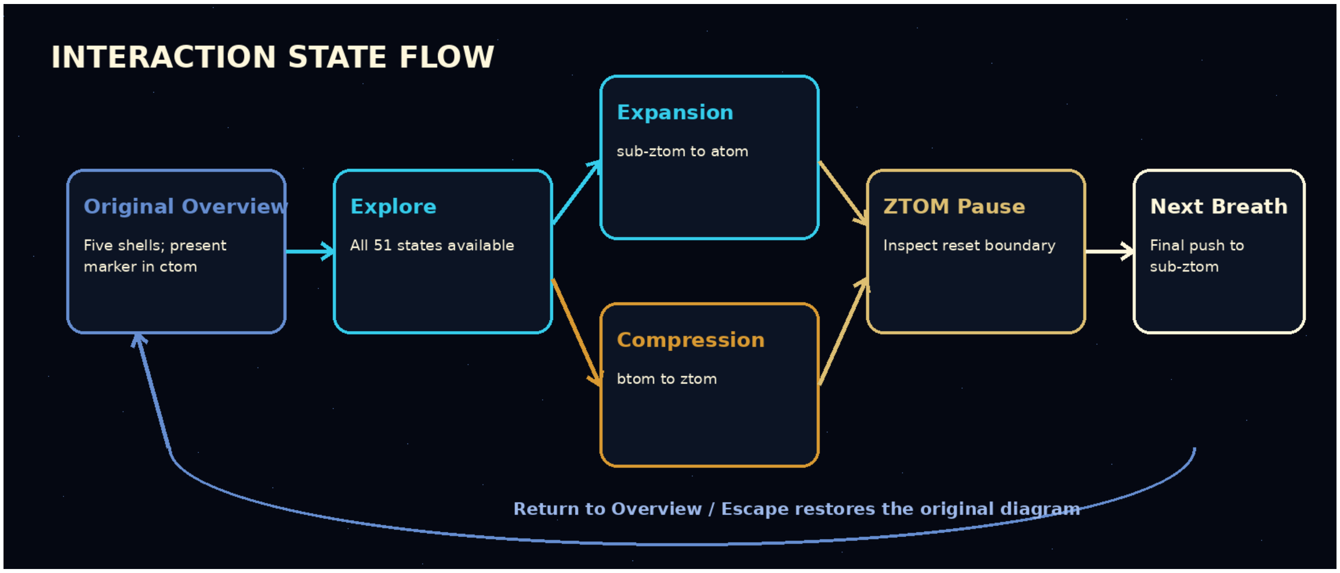

Figure 1: Required interaction-state flow. Returning to overview restores the unchanged original diagram.

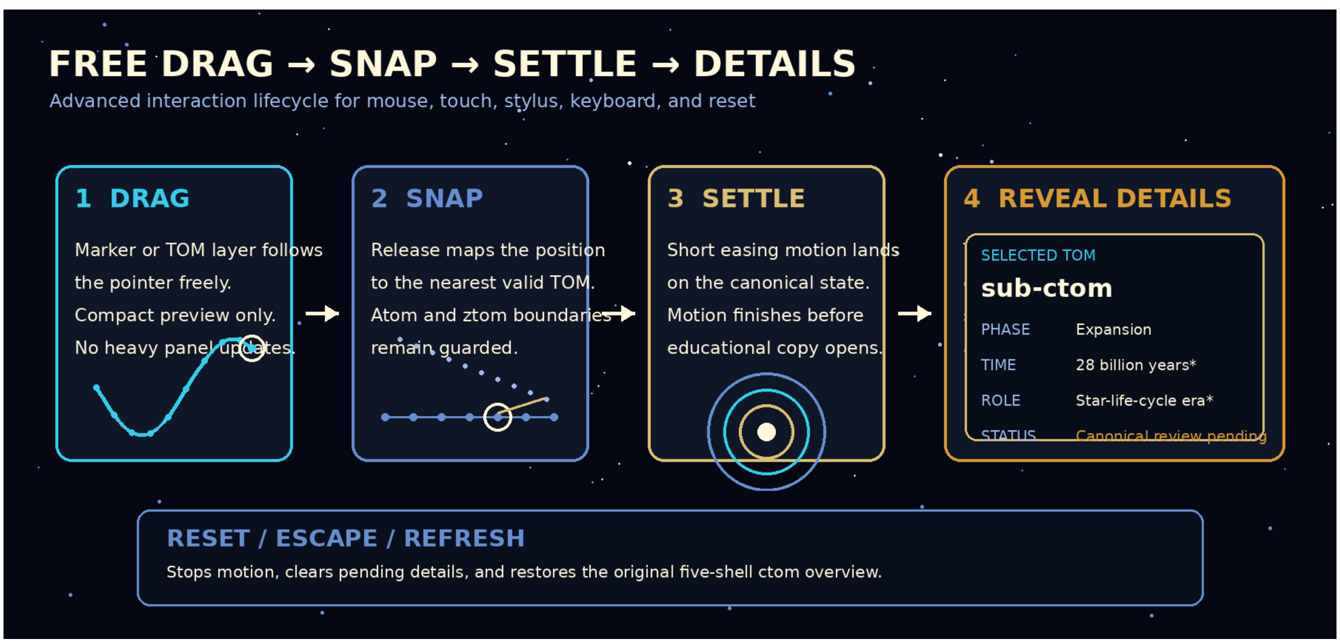

Figure 2: Version 1.1 motion lifecycle. Free dragging remains visually responsive, release snaps to a valid TOM, and the full educational panel appears only after settling.

3.3 Free Drag and Settle-to-Details

Exploration mode shall permit free-feeling movement without allowing an undefined final TOM state:

- the visitor may press a draggable simulated Cosmic State marker, a selected TOM-layer handle, or the cycle-position handle;
- the object follows the pointer continuously across the permitted visualization field rather than jumping immediately between labels;
- a compact live preview shows only the nearest TOM name, phase, and sequence position while movement is active;
- the full educational panel remains collapsed or visually subdued during active movement;
- on release, the object maps to the nearest valid TOM, respects atom and ztom boundary guards, and eases into its canonical position;
- after the settling animation and a short stability window, the complete details panel appears;
- beginning another drag immediately dismisses stale detail emphasis and starts a new motion cycle.

A short movement threshold may distinguish a click from a drag. A long-press threshold must not be required for mouse users and must never be the only way to enter exploration.

3.4 Persistent Exploration

A click, tap, Enter, or Space activation opens a persistent explorer. The visitor may drag repeatedly, read each settled TOM, select any TOM directly, use Previous and Next, use the native range control, open the full layer drawer, reset the explorer, or return to overview. Releasing a drag does not close persistent exploration.

3.5 Temporary Preview Option

A later implementation may support a temporary press-and-hold preview that returns to overview on release, but this is secondary. Version 1.1 prioritizes persistent exploration because visitors need time to read the details revealed after each movement.

3.6 Phase boundary behavior

At atom, the explorer pauses and announces the end of expansion. It shall not silently skip into btom. The interface then offers a clear continuation into compression.

At ztom, the explorer pauses again. It shall show the reset boundary and require a deliberate **Begin the Next Cosmic Breath** action for the final inward transition to sub-ztom.

3.7 Return behavior

Return to Overview, the visible Reset Explorer button, Escape, page reload, or an authorized reset shall:

- stop all motion;
- clear transient selections;

- collapse the detailed timeline and layer drawer;
- restore the present Observable Universe marker within ctom;
- restore the original five-shell visual state;
- clear pointer capture, pending animation frames, settle timers, and quality-session state;
- avoid storing state persistently.

Reset Explorer returns to the original ctom overview without reloading the page. A browser refresh produces the same default because the feature shall not use local storage, session storage, cookies, URL state, or server persistence.

4 Part 4 - Information Architecture and Educational Detail

4.1 Selected TOM panel

Every selected state shall be able to display:

- TOM label and normalized ID;
- phase: expansion, boundary, compression, reset, or overview;
- sequence position within the phase and complete cycle;
- duration value and duration approval status;
- quantum-state notation and notation approval status;
- CU description;
- previous and next states;
- distance to atom, ztom, or the next sub-ztom seed;
- editorial classification;
- source disclosure and unresolved notes when applicable.

4.2 Detail reveal timing

The panel shall distinguish motion from reading:

1. During active drag, show only a compact preview that can update at animation-frame cadence without reflowing the full panel.
2. On pointer release, compute the snap target once, enter Settling, and suppress the full panel until the marker is stable.
3. Reveal the complete panel after the settle animation plus a configurable stability delay in the approximate range of 180–250 milliseconds.

4. If a new pointer, keyboard, button, drawer, or slider action begins before reveal, cancel the pending reveal and follow the newest state.
5. Once visible, details remain until another movement, direct selection, Reset, Return to Overview, Escape, or refresh.

The delay is an interaction-design constant, not a CU-Time value. It may be tuned through browser testing without altering the TOM ledger.

4.3 Collapsed versus complete layer display

The overview may use an ellipsis to summarize the sequence, for example:

Collapsed sequence copy

```
Expansion:   sub-ztom -> sub-ytom -> sub-xtom -> ... -> sub-btom -> atom
Compression: btom -> ctom -> dtom -> ... -> ytom -> ztom
```

The locked explorer shall expose all 51 source-ledger states. The ellipsis must never make a state unreachable.

4.4 Full layer drawer

The All TOM Layers drawer shall group entries by phase, preserve deterministic order, identify the selected state, and permit direct selection. It should remain usable at 320px without horizontal overflow.

4.5 Source and interpretation disclosure

The panel shall include a disclosure similar to:

> **Source status.** TOM order and provisional descriptions derive from the supplied CU cycle ledger. Duration and notation values marked "review pending" have not yet passed the canonical TOM ledger review. The visualization is a CU theoretical model, not an empirical cosmological map.

5 Part 5 - Graphics, Color, Motion, and Responsive Design

5.1 Visual objective

The explorer should feel like a NASA-style scientific visualization combined with an academic research interface and restrained cosmic-documentary motion. It must remain credible, readable, and technically feasible in SVG, CSS, and lightweight JavaScript.

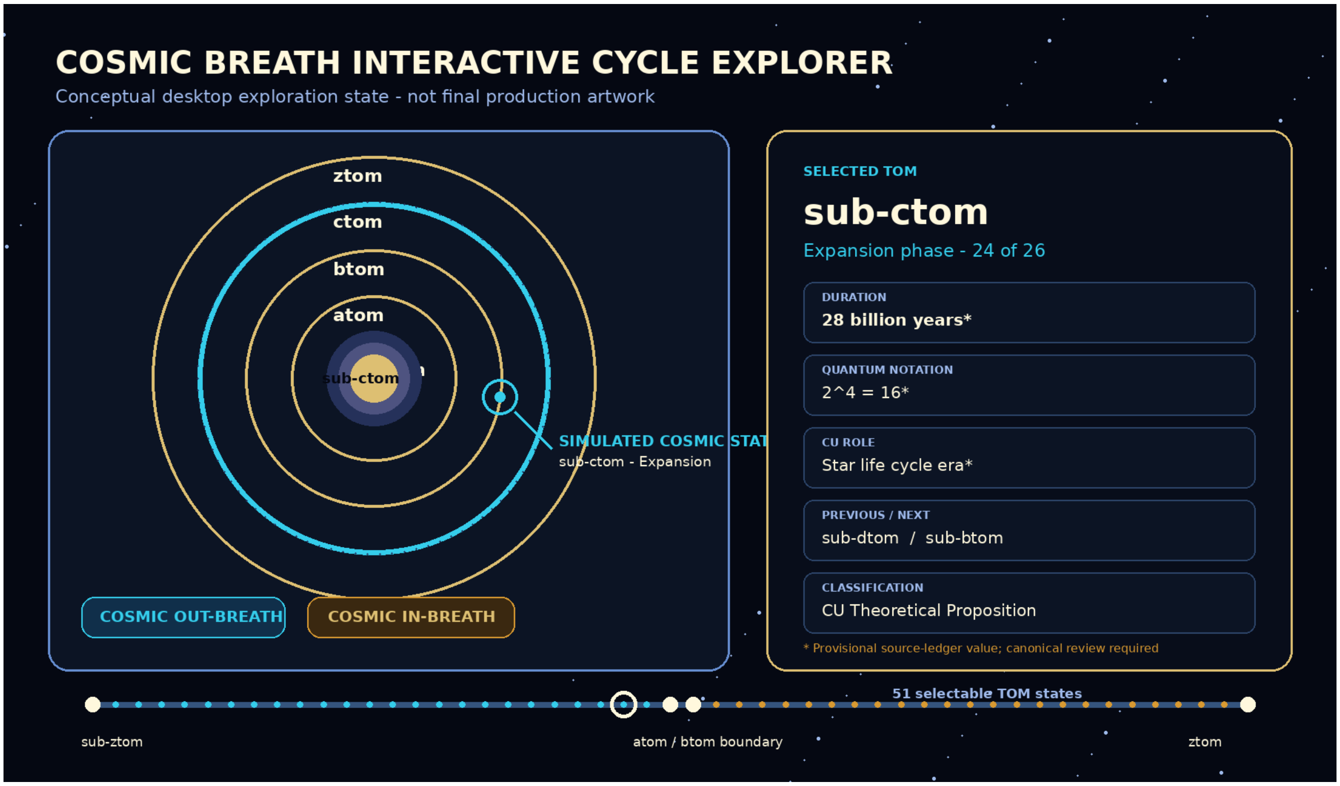

Figure 3: Conceptual desktop exploration state. Provisional values are visibly marked and the traveling object is labeled as a simulated cosmic state.

5.2 Phase palette

State	Primary color	Use
Overview	White-gold and cyan	Existing structural labels and present ctom marker.
Expansion	Cyan and translucent blue	Outward traces, selected shell, expansion progress, cool halo.
Atom boundary	White-gold	Pause, phase-turning emphasis, high-contrast transition.
Compression	White-gold and amber	Inward traces, increased density, compression progress.
ZTOM reset	Concentrated white-gold	Final boundary, contraction pulse, reset action.
Unresolved data	Amber	Review-pending values and source warnings.

5.3 Hybrid rendering architecture

Use a layered renderer rather than placing every visual effect in the DOM:

1. a semantic HTML control and details layer;
2. an SVG foreground for shells, TOM geometry, marker, labels, leader lines, focus rings, hit regions, and progress paths;

3. an optional Canvas background for dense particles, field traces, star drift, and phase atmosphere;
4. CSS transforms and opacity for GPU-friendly movement and fades;
5. a static SVG-only fallback when Canvas initialization fails or the selected quality profile disables it.

Canvas is decorative. TOM identity, focus, selected state, source status, and educational meaning must remain in semantic HTML/SVG. No visitor may lose information because Canvas, animation, or advanced effects are unavailable.

5.4 SVG foreground layers

Recommended SVG rendering order:

1. structural shell fills;
2. phase path and progress arc;
3. shell boundaries;
4. selected-state halo;
5. simulated Cosmic State marker and drag ghost;
6. labels and leader lines;
7. focus, pointer hit regions, and snap-target indication.

5.5 Motion rules

- Active drag follows the pointer with no decorative easing that creates a feeling of resistance or delay.
- Release uses a short, interruptible snap animation to the nearest canonical TOM.
- Expansion uses outward light movement, cool cyan flow, and gentle widening.
- Compression uses inward traces, white-gold/amber concentration, and increasing visual density.
- ZTOM uses a brief pause before a deliberate contraction into the next sub-ztom seed.
- UI animation duration shall not literally scale to trillions of years.
- Time magnitude is communicated by text and an optional logarithmic comparison graphic, not by making the visitor wait proportionally.
- `prefers-reduced-motion: reduce` removes nonessential travel, particle, pulse, parallax, blur interpolation, and camera effects while preserving immediate snap and state changes.
- No flashing pattern may exceed accepted accessibility thresholds.

5.6 Advanced visual effects

Subject to performance and accessibility gates, the explorer may add layered star fields, low-amplitude parallax, radial scattering, orbit traces, expansion wavefronts, compression streams, subtle lens-like distortion, phase-dependent density, and a concentrated reset pulse. Effects must support the selected TOM rather than obscure labels or simulate unsupported empirical measurements.

5.7 Adaptive visual quality and performance budget

The visual system shall expose or internally select one of four profiles: **Auto**, **High**, **Balanced**, and **Reduced**. Auto is the default. Reduced follows reduced-motion preferences and disables nonessential continuous animation.

Project performance targets, not universal guarantees, are:

- pointer movement reflected on the next available animation frame;
- approximately 60 frames per second on capable desktop hardware and a stable 30–60 frames per second on ordinary mobile hardware;
- no detail-panel reconstruction on every pointer event;
- no avoidable main-thread task longer than 50 milliseconds during normal drag;
- Canvas device-pixel ratio capped to a tested value, normally no greater than 2;
- particle and trace counts reduced for narrow screens and lower quality profiles;
- animation paused when the document is hidden or the explorer is outside the viewport;
- rendering performed only while dirty, dragging, settling, or intentionally animating.

Implementation should use Pointer Events, pointer capture, `requestAnimationFrame`, transform/opacity updates, cached geometry, and a single state source. Avoid layout-triggering per-frame writes, DOM nodes for individual particles, synchronous storage, and unbounded blur/filter effects.

5.8 Responsive layouts

Width	Required layout
1440px	Split visualization and detail panel; full controls and visible phase track.
1024px	Balanced split or compact two-column layout with no clipped labels.
768px	Stacked or narrow split; drawer and controls remain immediately reachable.
390px	Diagram above details; native slider and buttons become primary controls.
320px	Single column; no horizontal overflow; condensed labels; full list remains readable.

5.9 Annotated planning reference

6 Part 6 - Canonical TOM Ledger Review Gate

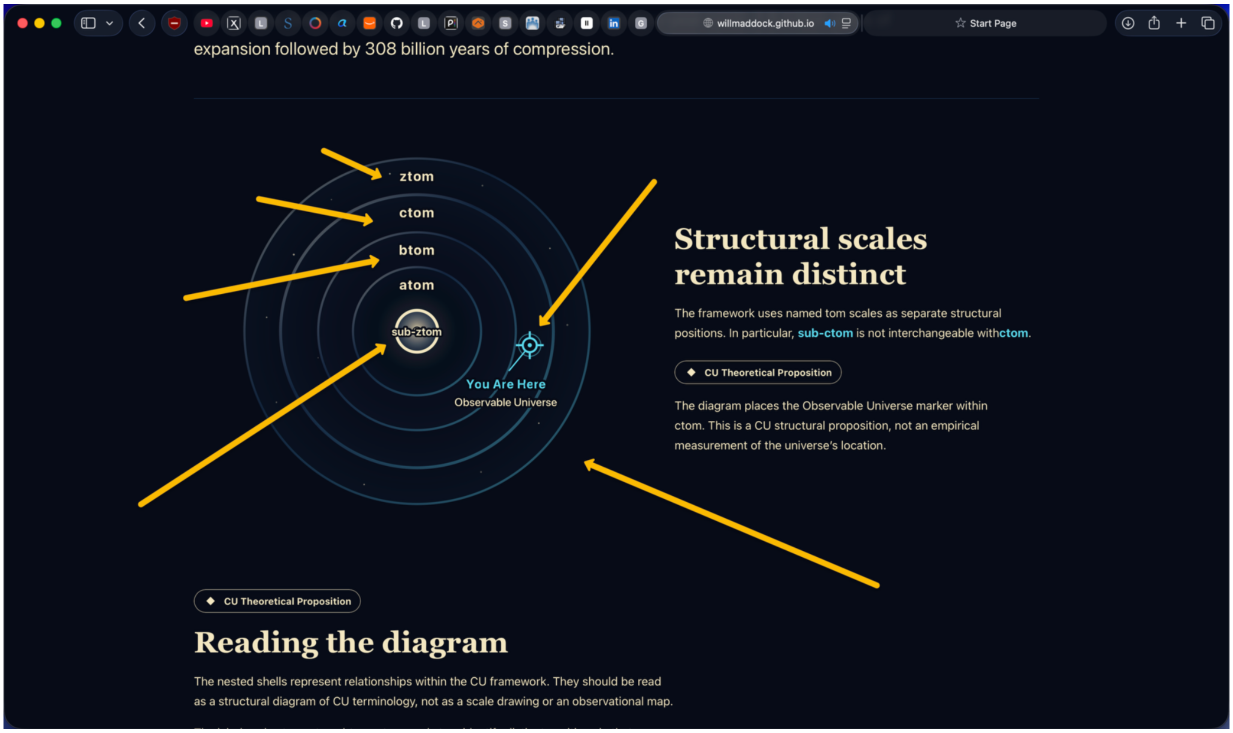

Figure 4: User-provided planning screenshot. Yellow arrows are annotations identifying desired interaction targets and are not final interface artwork.

6.1 What the uploaded sources establish

The full-cycle calculation source supplies a 26-state expansion sequence, a 25-state compression sequence, phase totals of approximately 2.8 trillion and 308 billion years, provisional quantum notation, provisional durations, and descriptive labels. The breathing-cycle source supplies the narrative of expansion, compression, ztom influence, and sub-ztom as the seed for another cycle. The separate time-calculation source supplies formulas and examples for exponential and tetration-based durations.

6.2 Why a review gate is mandatory

The uploaded sources are not fully consistent. Examples include:

- the time-calculation examples state arithmetic results that do not follow directly from the written formula for sub-btom and sub-ctom;
- several middle-level durations differ between the full-cycle ledger and the separate formula table;
- the tetration index mapping differs between documents;
- one source uses atom as a 2.8-trillion-year phase endpoint while another note maps ATOM to Planck time;
- source narratives vary on the direction and terminology of the reset transition.

Do Not Continue Until Resolved

Do not silently select one conflicting value, "fix" a source inside production code, or present an unresolved duration as canonical. The implementation may support the full sequence and review-status fields before final values are approved.

6.3 Required review outputs

The canonical TOM ledger review shall produce:

1. one ordered expansion array and one ordered compression array;
2. one approved display name and normalized ID per state;
3. approved quantum notation or an explicit "not shown" decision;
4. approved duration display text and machine-readable magnitude where applicable;
5. phase totals and boundary semantics;
6. source provenance per field;
7. an explicit decision for every detected discrepancy;
8. a versioned ledger identifier and SHA-256 digest.

6.4 Typed approval fields

Required data-status model

```
type ReviewStatus = 'approved' | 'provisional' | 'withheld';

type TomPhase =
  | 'expansion'
  | 'expansion-boundary'
  | 'compression'
  | 'reset-boundary';

interface CosmicBreathTomState {
  id: string;
  label: string;
  phase: TomPhase;
  cycleIndex: number;
  phaseIndex: number;
  durationText?: string;
  durationStatus: ReviewStatus;
  quantumNotation?: string;
  notationStatus: ReviewStatus;
  description: string;
  classification: 'cu-theoretical-proposition';
  sourceNotes: readonly string[];
```

```
}
```

6.5 Approval rule for production display

An approved implementation may show provisional values only when the UI labels them visibly as provisional and the implementation authorization explicitly permits that behavior. The preferred release gate is that all publicly displayed duration and notation values are approved.

7 Part 7 - Technical Architecture and File Responsibilities

7.1 Architecture decision

Because the existing static diagram is shared with the homepage, version 1.1 shall add a route-specific explorer rather than converting the shared diagram into an always-interactive component. The route-specific architecture also isolates the advanced Canvas, drag, snap, and quality-management code from the homepage.

7.2 Smallest safe file set

Path	Responsibility
website/src/pages/cosmic-breath.astro	Replace only the page diagram slot with the route-specific explorer and preserve current explanatory sections.
website/src/components/CosmicBreathCycleExplorer.astro	Render overview fallback, interactive SVG, details, controls, drawer, scoped script, and component CSS.
website/src/lib/cosmicBreath/cycle-ledger.ts	Typed ordered TOM registry, provenance fields, review statuses, and lookup helpers.
website/src/lib/cosmicBreath/cycle-state.ts	Pure navigation, mode, boundary, marker-label, reset, drag lifecycle, settle, and detail-visibility state functions.
website/src/lib/cosmicBreath/cycle-drag.ts	Pure coordinate normalization, nearest-TOM mapping, drag threshold, snap-target, interruption, and settle helpers.
website/src/lib/cosmicBreath/cycle-quality.ts	Typed Auto/High/Balanced/Reduced profiles, particle budgets, device-pixel-ratio caps, and reduced-motion decisions.
website/src/lib/cosmicBreath/__tests__/cycle-ledger.test.ts	Registry order, uniqueness, phase, boundary, source, and review-status tests.
website/src/lib/cosmicBreath/__tests__/cycle-state.test.ts	Previous/next, jump, overview, details timing, reset, ctom marker, atom pause, ztom pause, and next-breath reset tests.
website/src/lib/cosmicBreath/__tests__/cycle-drag.test.ts	Coordinate mapping, free drag, nearest-state snapping, clamping, interruption, and settle tests.

Path	Responsibility
website/src/lib/cosmicBreath/__tests__/cycle-quality.test.ts	Quality-profile, reduced-motion, particle-budget, and device-pixel-ratio tests.

7.3 Protected existing files

The following should remain unchanged unless a later inspection proves a bounded need:

- website/src/components/CosmicBreathDiagram.astro;
- website/src/components/Hero.astro;
- CU-Time libraries and tests;
- CUCII registries, fixtures, builders, and tests;
- site navigation, routes, dependencies, Astro base configuration, and package versions.

7.4 Progressive enhancement

The server-rendered output shall include the original static figure and essential explanatory content. Interactive controls shall become available only after successful browser initialization. Failure to initialize must leave a useful static page rather than an empty shell.

7.5 Component scoping

All selectors and event handling must be scoped to the explorer root. Do not rely on global IDs or unscoped `document.querySelector` calls that break when multiple figures appear on one page.

8 Part 8 - Interaction Engineering

8.1 Canonical controls

The feature shall provide:

- Explore Full Cosmic Breath;
- draggable simulated Cosmic State marker and selected TOM-layer handle;
- Previous and Next;
- Play/Pause only if separately approved;
- a native range input representing indices 0 through 50;
- All TOM Layers disclosure;
- Reset Explorer;
- Return to Overview;

- Begin the Next Cosmic Breath at ztom;
- Visual Quality: Auto, High, Balanced, and Reduced, when the quality selector is included.

8.2 Draggable subjects and free movement

The visitor may drag the simulated Cosmic State marker, the current TOM-layer handle, or the dedicated cycle handle. The selected visual should follow two-dimensional pointer movement within a bounded exploration field. The implementation may use radial distance, angular phase regions, or a curved cycle path to determine the nearest TOM, but the user-facing movement should feel continuous rather than like a coarse slider.

Dragging a shell shall not permanently deform the CU structure. The shell may translate, pulse, or preview depth during movement, but its released state must resolve to a canonical TOM representation. The original overview geometry remains a protected reset target.

8.3 Pointer lifecycle

Use Pointer Events so one implementation supports mouse, finger, and stylus:

1. on pointer down, verify a permitted primary pointer and record origin;
2. after the movement threshold, enter Dragging and capture the pointer;
3. on each animation frame, consume the latest pointer coordinates, update the visual transform, and compute the nearest snap preview;
4. on pointer up, pointer cancel, or lost capture, resolve one snap target and enter Settling;
5. after settling, reveal details unless the action was interrupted or reset;
6. always release capture and cancel obsolete frames/timers during Reset, Escape, navigation, or component teardown.

Touch behavior must preserve normal page scrolling outside the designated drag surface. Apply `touch-action` only to the bounded interactive region and only as narrowly as required.

8.4 Snap and settle behavior

The nearest-state algorithm shall be deterministic, tested, and independent of frame rate. It shall consider the permitted phase region and clamp at atom and ztom until explicit boundary continuation. A snap animation should be brief, interruptible, and based on cached geometry. Details reveal only after the settle completion signal and stability delay.

If the visitor drags again during settling, cancel the prior animation and continue from the currently rendered position. If the visitor chooses a direct TOM, button, slider, or drawer item, the newest explicit command wins.

8.5 Reset semantics

Reset Explorer shall synchronously stop active drag, release pointer capture, cancel animation frames and settle timers, clear selected details, restore Auto quality unless the final implementation explicitly preserves a user-selected session preference, and return the diagram to the original five-shell ctom overview. Page refresh must produce the same visual and semantic state without reading persisted data.

8.6 Native slider synchronization

The custom marker and native range input share one state source. The range input provides a robust keyboard and assistive-technology fallback. Update `aria-valuetext` with a phrase such as "sub-ctom, expansion state 24 of 26, cycle state 24 of 51."

8.7 Keyboard behavior

Key	Required behavior
Left / Down	Previous TOM state.
Right / Up	Next TOM state.
Home	First expansion state: sub-ztom.
End	Final compression state: ztom.
Page Up	Previous major milestone.
Page Down	Next major milestone.
Enter / Space	Open explorer or activate focused control.
Escape	Return to the original overview.

8.8 Marker identity rules

Context	Visible identity
Overview	"You Are Here - Observable Universe - within ctom."
Exploring ctom	"Observable Universe / present ctom reference" plus simulation disclosure.
Exploring any other TOM	"Simulated Cosmic State" and the selected TOM.
At ztom	"Universal State - ztom reset boundary."
After final push	"New Cosmic Seed - sub-ztom - next breath."

8.9 Boundary guards

The state controller shall not wrap directly from atom to btom or from ztom to sub-ztom without an explicit boundary action. Accidental drag overshoot should clamp at the boundary until the visitor confirms continuation.

9 Part 9 - Accessibility, Usability, and Content Safety

9.1 Accessibility baseline

- No drag-only operation.
- Visible focus for every control.
- Native buttons and native range input wherever possible.
- Minimum touch targets consistent with the existing site standard.
- Text labels for all color-coded phases.

- A polite live region for user-initiated selection changes.
- No live announcement on every pixel of a drag; announce the final snapped TOM and details reveal only.
- Reduced-motion mode.
- Static no-JavaScript fallback.
- A non-drag control path with buttons, range input, and full layer drawer.
- No focus movement caused merely by pointer settling; the visitor retains control of focus.
- Meaningful SVG title and description for overview and exploration states.
- The full layer drawer remains navigable in document order.

9.2 Cognitive load

The interface shall not display 51 labels on top of the concentric diagram. It should show the selected state, nearby context, a phase track, and a complete drawer. Educational copy uses short sections and expandable source details.

9.3 Reality and claims boundary

The feature shall consistently distinguish CU theoretical propositions from empirical cosmology. It shall not claim that the interaction moves the physical Observable Universe, that the diagram is observationally measured, or that the browser simulation validates CU mathematics.

9.4 Privacy and persistence

The feature requires no account, analytics, telemetry, backend, storage, cookies, or persisted explorer state. Any future persistence or analytics requires separate approval.

10 Part 10 - Automated Tests, Browser Matrix, and Acceptance

10.1 Focused automated tests

The new tests shall verify at minimum:

- 26 expansion and 25 compression entries when the approved ledger retains the supplied sequence;
- exactly 51 unique IDs and cycle indices;
- deterministic first, atom boundary, btom boundary, ctom, ztom, and reset transitions;
- no sub-ctom/ctom conflation in IDs or descriptions;
- present Observable Universe identity only in the overview/ctom context;
- provisional or withheld values cannot be rendered as approved;
- Previous/Next and drag snapping clamp at phase boundaries until explicit continuation;

- free pointer coordinates map deterministically to the nearest valid TOM;
- details remain suppressed while dragging and appear after settle completion plus the configured stability delay;
- a new movement cancels stale settle and detail timers;
- Reset Explorer, Escape, and initialization clear transient state and restore the original ctom overview;
- refresh behavior requires no persisted state;
- quality profiles apply bounded particle counts, device-pixel-ratio caps, and reduced-motion rules;
- full source provenance and classification on every entry.

10.2 Full project gate

After implementation changes, run:

Final local verification

```
cd /Users/cosmos/Documents/GitHub/CosmicUniversalismStatement/website || exit 1

npm test
npm run build

cd ..
git diff --check
git status --short --branch
git diff --stat
git ls-files --others --exclude-standard | sort
```

10.3 Manual browser matrix

Test at 1440, 1024, 768, 390, and 320 pixels. At each width verify overview, entry, free marker drag, TOM-layer drag, compact motion preview, snap target, settle interruption, post-settle detail reveal, repeated moves, Reset Explorer, refresh reset, range input, Previous/Next, full drawer, atom boundary, compression entry, ztom pause, next-breath reset, Return to Overview, focus, announcements, Auto/High/Balanced/Reduced quality, reduced motion, no-JavaScript fallback, page scrolling outside the drag surface, and browser console.

10.4 Acceptance criteria

The feature is acceptable only when:

1. the original overview remains visually and semantically intact;
2. all approved TOM states are reachable without dragging;
3. free pointer movement feels continuous while released positions resolve to the same canonical state order used by keyboard and buttons;

4. full details do not churn during drag and appear reliably after settling;
5. repeated drag, release, read, and move cycles do not leak timers, pointer capture, or stale details;
6. Reset Explorer and refresh restore the exact original overview;
7. atom and ztom pauses cannot be skipped accidentally;
8. the present Observable Universe marker is not misrepresented outside ctom;
9. review-pending values are not presented as canonical;
10. the homepage remains unchanged;
11. performance meets the approved device/quality targets without blocking basic interaction;
12. all tests and the production build pass;
13. no console errors, overflow, clipping, inaccessible controls, broken base-path assets, or uncontrolled continuous rendering remain.

11 Part 11 - Controlled Implementation, Git, Release, and Recovery

11.1 Pre-implementation preflight

Verify the live repository before branch creation

```
cd /Users/cosmos/Documents/GitHub/CosmicUniversalismStatement || exit 1

git rev-parse --show-toplevel
git remote get-url origin
git branch --show-current
git rev-parse --short HEAD
git status --short --branch
git tag --points-at HEAD
```

Do not assume `d426b78` remains current. Compare the future output with the authorized starting baseline.

11.2 Proposed branch and release

Proposed feature branch	`feature/cosmic-breath-cycle-explorer`
Proposed commit title	Add interactive Cosmic Breath cycle explorer
Proposed release tag	`website-v1.4-cosmic-breath-explorer-release`
Recovery point	`website-v1.3-aurelius-release`

These are proposals, not authorizations or records of completed Git actions.

11.3 Bounded implementation authorization template

Copy-and-Paste Prompt: Authorize the Cycle Explorer Implementation

```
The inspection plan and canonical TOM ledger are approved with the
following exact scope.

OBJECTIVE:
Implement the route-specific Cosmic Breath Interactive Cycle Explorer.

AUTHORIZED FILES:
website/src/pages/cosmic-breath.astro
website/src/components/CosmicBreathCycleExplorer.astro
website/src/lib/cosmicBreath/cycle-ledger.ts
website/src/lib/cosmicBreath/cycle-state.ts
website/src/lib/cosmicBreath/__tests__/cycle-ledger.test.ts
website/src/lib/cosmicBreath/__tests__/cycle-state.test.ts

REQUIRED BEHAVIOR:
- Preserve the unchanged overview and homepage.
- Expose all approved TOM states.
- Support pointer, touch, stylus, buttons, range input, and keyboard.
- Show approved time, notation, phase, details, provenance, and status.
- Pause at atom and ztom.
- Require a deliberate ztom-to-sub-ztom next-breath action.
- Restore overview on Escape or Return to Overview.

PRESERVE:
- main branch
- CU-Time mathematics and approved source URLs
- current CosmicBreathDiagram.astro and Hero.astro unless separately authorized
- base-path behavior
- existing tests and routes
- CUCII claims boundary

Run npm test, npm run build, git diff --check, git status
--short --branch, git diff --stat, and the full browser matrix.

Do not add dependencies or perform any Git-changing or deployment action.
```

11.4 Release gate

Commit, push, merge, tag, release, and deployment require separate explicit authorization after final acceptance. A successful implementation test does not grant release permission.

11.5 Recovery

Use the prior stable tag as a recovery point. Do not rewrite published history. A recovery inspection should occur on a newly created branch, not by editing a detached tag.

12 Part 12 - Implementation Phases

12.1 Phase 0 - Canonical ledger approval

Resolve all duration, notation, ordering, and reset semantics. Produce a versioned data artifact and digest.

12.2 Phase 1 - Typed registry and pure state engine

Implement the approved registry, lookup helpers, boundary helpers, mode model, and tests before creating complex SVG interaction.

12.3 Phase 2 - Static route-specific shell

Add the new component with the existing static overview as a progressive fallback. Confirm no homepage change and no JavaScript dependency for basic comprehension.

12.4 Phase 3 - Native controls and details

Add buttons, range input, full layer drawer, compact motion preview, full details panel, Reset Explorer, and live region. Complete keyboard and reset behavior before pointer dragging.

12.5 Phase 4 - Pure drag, snap, and settle engine

Implement and test coordinate normalization, movement thresholds, nearest-TOM mapping, phase clamping, interruption, settle completion, and detail timing before connecting high-fidelity graphics.

12.6 Phase 5 - Pointer and SVG interaction

Add hit regions, pointer capture, free-feeling marker and TOM-layer travel, snap preview, selected shell styling, phase progress, boundary guards, and repeated move-to-details cycles.

12.7 Phase 6 - Canvas atmosphere and adaptive quality

Add optional particle fields, traces, halos, parallax, and reset transition behind the semantic SVG. Implement Auto/High/Balanced/Reduced profiles, frame budgeting, visibility pausing, device-pixel-ratio caps, and reduced motion.

12.8 Phase 7 - Full verification

Run focused tests, full tests, production build, drag performance sampling, browser matrix, accessibility review, console review, refresh/reset review, and Git inspection. Correct only reproducible defects within scope.

13 Part 13 - Risk Register and Decision Rules

Risk	Failure mode	Required control
Ledger conflict	Incorrect time presented as canonical.	Canonical review, per-field status, provenance tests.

Risk	Failure mode	Required control
Shared component	Homepage changes unintentionally.	Route-specific explorer; protect Hero and static diagram.
Drag-only control	Keyboard and assistive-tech exclusion.	Native range, buttons, drawer, focus, announcements.
Free-drag ambiguity	Arbitrary released position implies a nonexistent TOM.	Continuous visual movement plus deterministic canonical snap.
Stale details	Prior TOM details remain after a new drag or interrupted settle.	Single state source; cancel obsolete timers and reveals.
Touch conflict	Drag surface prevents normal mobile page scrolling.	Bounded hit surface and narrowly scoped touch-action.
Fifty-one labels	Visual clutter and unreadability.	Selected-state view plus complete drawer.
Boundary overshoot	atom or ztom skipped.	Clamp and explicit continuation actions.
Marker semantics	Present universe shown in every TOM.	Simulated Cosmic State identity outside ctom overview.
Motion overload	Disorientation or accessibility failure.	Reduced motion, restrained cadence, no flashing.
Mobile overflow	Timeline or panel unusable at 320px.	Stacked layout and native controls.
Performance	Excessive particles, filters, canvas resolution, or per-event DOM work.	Adaptive quality, capped DPR, bounded particles, requestAnimationFrame, cached geometry, visibility pause, and transform-based animation.
Source drift	Manual baseline becomes stale.	Re-preflight and live-file inspection before edits.

14 Appendix A - Feature Planning Worksheet

Field	Approved manual entry
Feature name	Cosmic Breath Interactive Cycle Explorer.
User outcome	Freely drag a simulated state or TOM layer, see compact live TOM previews, release to snap and reveal complete educational details, repeat across all expansion/compression states through ztom and next-breath sub-ztom, and reset to the original overview.
Baseline	Recorded live integration `d426b78`; latest tagged recovery `5a56085`.
Feature branch	Proposed `feature/cosmic-breath-cycle-explorer`; not created.
Routes	Existing /CosmicUniversalismStatement/cosmic-breath/ only.
Authorized files	To be approved later using the Part 7 file set.
Protected behavior	Homepage, static diagram, ctom marker meaning, CU-Time, CUCII, routes, base path, dependencies, existing tests.

Data model	Typed TOM registry with provenance and approval statuses; pure cycle state engine.
Accessibility	Native slider, buttons, keyboard, focus, live status, reduced motion, no-JS fallback.
Tests	Two focused test files plus full project suite and build.
Browser matrix	1440, 1024, 768, 390, 320px.
Commit	Proposed “Add interactive Cosmic Breath cycle explorer.”
Release	Proposed `website-v1.4-cosmic-breath-explorer-release`.
Recovery	`website-v1.3-aurelius-release`.

15 Appendix B - Provisional Full-Cycle Source Ledger

Do Not Continue Until Resolved

The following tables reproduce the supplied full-cycle source ledger for planning. “Provisional” means the value is not approved for unqualified production display. This appendix does not reconcile the conflicts described in Part 6.

15.1 Expansion source ledger - 26 states

#	TOM	Quantum	Duration	CU description	Status
1	sub-ztom	2^^65,536	1 second	Final compression breath	Provisional
2	sub-ytom	2^^40	2.704e-8 seconds	Recursive core shell	Provisional
3	sub-xtom	2^^30	2.704e-7 seconds	Final ethical cap	Provisional
4	sub-wtom	2^^22	2.704e-6 seconds	Quantum firewall shell	Provisional
5	sub-vtom	2^^21	2.704e-5 seconds	Symbolic/ethical lock	Provisional
6	sub-utom	2^^20	0.0002704 seconds	Cosmic checksum layer	Provisional
7	sub-ttom	2^^19	0.002704 seconds	Holographic verification	Provisional
8	sub-stom	2^^18	0.02704 seconds	Ethical engagement	Provisional
9	sub-rtom	2^^17	0.2704 seconds	Entropy modulation	Provisional
10	sub-qtom	2^^16	2.704 seconds	Memory transfer	Provisional
11	sub-ptom	2^^15	27.04 seconds	Pre-reset bridge	Provisional
12	sub-otom	2^^14	4.506 minutes	Boundary stabilization	Provisional
13	sub-ntom	2^^13	45.06 minutes	Recursive feedback	Provisional
14	sub-mtom	2^^12	7.51 hours	Holographic projection	Provisional
15	sub-ltom	2^^11	3.1296 days	Pre-Big Bang state	Provisional
16	sub-ktom	2^^10	31.296 days	Quantum foam rebirth	Provisional
17	sub-jtom	2^^9	0.8547 years	Black hole age	Provisional
18	sub-itom	2^^8	8.547 years	Spacetime contraction begins	Provisional
19	sub-htom	2^^7	85.47 years	Heat death begins	Provisional
20	sub-gtom	2^^6	427.35 years	Quantum encoding phase	Provisional
21	sub-ftom	2^^5	4,273.5 years	Post-biological AI expansion	Provisional
22	sub-etom	2^^4	42,735 years	Alien/civilization stage	Provisional
23	sub-dtom	2^16 = 65,536	427,350 years	Planetary biosphere evolution	Provisional
24	sub-ctom	2^4 = 16	28 billion years	Star life cycle era	Provisional
25	sub-btom	2^2 = 4	280 billion years	Supercluster formation	Provisional
26	atom	2^1 = 2	2.8 trillion years	Start of compression	Provisional

15.2 Compression source ledger - 25 states

#	TOM	Quantum	Duration	CU description	Status
1	btom	2^2 = 4	280 billion years	Galactic evolution and contraction	Provisional
2	ctom	2^4 = 16	28 billion years	Final stellar formations	Provisional
3	dtom	2^16 = 65,536	427,350 years	Planetary collapse	Provisional
4	etom	2^^4	42,735 years	Human/civilization memory condensation	Provisional
5	ftom	2^^5	4,273.5 years	AI implosion stage	Provisional
6	gtom	2^^6	427.35 years	Consciousness holography	Provisional
7	htom	2^^7	85.47 years	Heat death approach	Provisional
8	itom	2^^8	8.547 years	Spacetime wrinkle forming	Provisional
9	jtom	2^^9	0.8547 years	Collapse threshold	Provisional
10	ktom	2^^10	31.296 days	Quantum fog closing	Provisional
11	ltom	2^^11	3.1296 days	Holographic reversal	Provisional
12	mtom	2^^12	7.51 hours	Time lattice inversion	Provisional
13	ntom	2^^13	45.06 minutes	Feedback end	Provisional
14	otom	2^^14	4.506 minutes	Cosmic null stabilization	Provisional
15	ptom	2^^15	27.04 seconds	Reset layering	Provisional
16	qtom	2^^16	2.704 seconds	Final memory imprint	Provisional
17	rtom	2^^17	0.2704 seconds	Entropy zero point	Provisional
18	stom	2^^18	0.02704 seconds	Ethical firewall gate	Provisional
19	ttom	2^^19	0.002704 seconds	Collapse checksum	Provisional
20	utom	2^^20	0.0002704 seconds	Closure sequence initiated	Provisional
21	vtom	2^^21	2.704e-5 seconds	Symbolic compression	Provisional
22	wtom	2^^22	2.704e-6 seconds	Recursive limit breach	Provisional
23	xtom	2^^30	2.704e-7 seconds	Divine fall-off shell	Provisional
24	ytom	2^^40	2.704e-8 seconds	Pre-ZTOM divine echo	Provisional
25	ztom	2^^65,536	1 second	ZTOM: full universal reset	Provisional

16 Appendix C - Canonical Ledger Review Worksheet

For each TOM state, record:

Field	Required decision
Normalized ID	Stable lowercase identifier and uniqueness proof.
Display label	Approved capitalization and hyphenation.
Phase and order	Expansion/compression/boundary classification and indices.
Quantum notation	Approved notation, accessibility text, or withheld decision.
Duration display	Approved human-readable value or withheld decision.
Machine magnitude	Unit and numeric representation if needed for comparison graphics.
Description	Approved educational copy and classification.
Source provenance	File, section/table, and revision date.
Conflict decision	Explanation of any difference from another source.
Reviewer approval	Name, date, ledger version, and digest.

17 Appendix D - Browser Review Script

1. Load the route with JavaScript enabled and confirm the unchanged overview.
2. Disable JavaScript and confirm the static diagram and explanatory copy remain.
3. Enter locked exploration by mouse, touch emulation, and keyboard.
4. Move to every major milestone and sample intermediate states.
5. Open All TOM Layers and directly select early, middle, and late entries.
6. Verify atom pause, compression continuation, ztom pause, and next-breath reset.
7. Press Escape and confirm exact overview restoration.
8. Repeat with reduced motion.
9. Repeat at 1440, 1024, 768, 390, and 320px.
10. Inspect focus order, accessible names, live announcements, and console output.

18 Appendix E - Source Provenance

Source	Use in this manual
Master Implementation and Operations Manual v1.0 PDF and LaTeX	Governance, current tagged release record, source priority, technology, workflow, test gate, LaTeX house style.
User Git preflight and history outputs	Live repository root, origin, branch, clean status, current HEAD, ancestry, and post-release documentation merge.
Pastedtext(8).txt	Current Cosmic Breath page and static diagram source inspection.
Pastedtext(9).txt	Homepage shared-component use, package/test configuration, interaction conventions, and project test manifest.
Cosmic_Breath_Calculation(4).md	Provisional 51-state order, durations, notations, descriptions, and phase totals.
Cosmic_Breathing_Cycle(4).md	Expansion/compression narrative and sub-ztom next-cycle seed interpretation.
Time_Calculation(2).md	Duration formulas, examples, symbolic-time framing, and detected mathematical conflicts.
cosmic-breath-annotated-planning-reference.png	User's visual interaction-target reference; not final artwork.
Conversation decisions through July 22, 2026	Approved feature outcome, all-layer exploration, movable simulated state, time/detail panel, graphics direction, reset and overview behavior.

19 Appendix F - Current-State Handoff Summary

Project current state

```
Repository: /Users/cosmos/Documents/GitHub/CosmicUniversalismStatement
Origin: https://github.com/willmaddock/CosmicUniversalismStatement.git
Protected branch: main
Stable integration branch: website
Recorded live HEAD: d426b78
Latest tagged application release: website-v1.3-aurelius-release
Tagged release commit: 5a56085
Working tree at inspection: clean and synchronized
Active feature branch: none
Manual package: created outside the repository
Website files changed: none
Git actions performed: none
Next gate: canonical TOM ledger review, then re-preflight and bounded implementation
    authorization
```

20 Appendix G - Revision Record

Version	Date	Author	Record
1.0	July 22, 2026	William Maddock	Initial controlled feature manual. Captures inspected live baseline, complete interaction specification, 51-state source ledger, graphics and accessibility system, canonical ledger review gate, implementation phases, test plan, and release proposal.
1.1	July 22, 2026	William Maddock	Advanced interaction revision. Adds free planar dragging of the simulated Cosmic State and TOM-layer handles; deterministic nearest-TOM snapping; interruptible settling; detail reveal only after motion stops; repeated drag/read cycles; visible reset and refresh semantics; hybrid SVG/Canvas rendering; adaptive visual quality; explicit performance budgets; expanded tests, browser review, risks, and implementation phases. Corrects cover-page paragraph separation in the executable LaTeX source and verifies the rendered title hierarchy.

Manual Status

This version 1.1 package was created under explicit authorization for manual revision only. No repository file, branch, commit, tag, release, or deployment was changed.